PROJECTS

MLB Postseason Prediction - ETSU - Honors Thesis

- Analyzed 1998–2024 MLB team-season data to identify statistical factors associated with postseason success.
- Evaluated multiple statistical and machine learning approaches, including PCA, clustering, logistic regression, LDA, decision trees, bagging, and Random Forest.
- Identified ERA+, Runs Allowed, OPS+, and Runs Scored among the most important variables for distinguishing postseason outcomes.
- Selected a Random Forest model to generate predictions for the 2025 MLB postseason.

LLM Consistency Experiment - UC Berkeley MIDS - Experimentation & Causal Inference

- Designed and conducted a controlled experiment investigating whether consistency-focused prompting affects the consistency of repeated LLM responses.
- Evaluated Gemini 2.5 Flash, GPT-4.1 Mini, and Gemma-3 across text and image-based tasks using repeated generations under controlled experimental conditions.
- Applied sentence-transformer embeddings (all-mpnet-base-v2) and cosine similarity to quantify semantic consistency across model responses.
- Analyzed treatment effects using regression models, interaction terms, and cluster-robust standard errors to evaluate differences across models and experimental conditions.
- Found that the consistency intervention reduced overall response consistency by approximately 20%, with effects varying substantially across models.

2026 FIFA World Cup Prediction - UC Berkeley MIDS - Applied Machine Learning

- Integrated 49,071 international match records and 13,130 FIFA ranking records, producing a modeling dataset of 25,691 observations.
- Engineered 15 pre-match features to predict international soccer match outcomes while avoiding information unavailable before matches.
- Compared logistic regression, neural networks, and gradient boosting models to evaluate predictive performance.
- Achieved 60.89% test accuracy with logistic regression and used model predictions to simulate the 2026 FIFA World Cup.

NFL Officiating Exploratory Data Analysis - UC Berkeley MIDS - Data Science Programming

- Conducted exploratory analysis of NFL penalties, referee assignments, replay reviews, and game-level data from 2015–2024.
- Integrated multiple datasets while cleaning inconsistent team, referee, date, and game identifiers.
- Investigated trends in penalty frequency, referee behavior, replay reviews, and officiating patterns across seasons and teams.
- Developed visualizations and statistical summaries to communicate trends and identify relationships within the data.

LEADERSHIP

ETSU Army ROTC
2021 - 2022

Contracted Cadet

- Served as team and squad leader during tactical field-training exercises.
- Trained to lead teams and make decisions in simulated high-stress environments.
- Led preparation and execution of morning physical training activities.